

Final Report :

Wave Angle & CFD on the Diamond-Shaped Airfoil

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1 Introduction

This experiment focuses on comparing the wave angle of 2D and 3D supersonic flow simulations, $M = 3$, over a diamond airfoil, 10° half-angle, using *OpenFOAM*. We show convergence of the 2D mesh and compare the pressure at the stagnation point to the theoretical stagnation pressure using oblique shock theory, normal shock relationships, and isentropic pressure relationships. Mesh refinement brings the value of β extremely close to theory. We will then analyze and compare the different wave angles to see if the *rhoCentralFoam* solver captures the influence of 3D effects on wave angle. This will be compared to experimental data, where the observed wave angle is usually steeper than the θ - β - M equation, derived from 2D inviscid Euler equations, would suggest.

2 Governing Equations

The flow is assumed to be inviscid, compressible, non-reacting and calorically perfect (isothermal) with a constant specific-heat ratio $\gamma = 1.4$. Under these hypotheses, the governing equations reduce to the *compressible Euler equations* complemented by the ideal-gas law. For completeness we also list the oblique-shock relations that form the theoretical reference for the diamond airfoil.

2.1 Three-dimensional Compressible Euler equations

From the assumptions of Euler equations, e.g., no body forces and inviscid and isothermal flow, the simplified form of the *Navier-Stokes Equations* in $\mathbb{R}^3$ reads:

- Continuity:

$$\frac{\partial \rho}{\partial t} + \rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = 0. \quad (1)$$

- X-Momentum:

$$\frac{\partial(\rho u)}{\partial t} + \rho \left(\frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} + \frac{\partial uw}{\partial z} \right) = -\frac{\partial p}{\partial x}. \quad (2)$$

- Y-Momentum:

$$\frac{\partial(\rho v)}{\partial t} + \rho \left(\frac{\partial vu}{\partial x} + \frac{\partial v^2}{\partial y} + \frac{\partial vw}{\partial z} \right) = -\frac{\partial p}{\partial y}. \quad (3)$$

- Z-Momentum:

$$\frac{\partial(\rho w)}{\partial t} + \rho \left(\frac{\partial wu}{\partial x} + \frac{\partial wv}{\partial y} + \frac{\partial w^2}{\partial z} \right) = -\frac{\partial p}{\partial z}. \quad (4)$$

- Energy:

$$\frac{\partial E}{\partial t} + E \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = -p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right), \quad (5)$$

Thus, the conservative state vector is

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ E \end{bmatrix}, \quad E = \rho \left(e + \frac{1}{2}(u^2 + v^2 + w^2) \right).$$

And the flux vectors, using $\mathbf{F}, \mathbf{G}, \mathbf{H}$ to denote fluxes in the x, y and z directions, respectively, are

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ (E + p)u \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ \rho vw \\ (E + p)v \end{bmatrix}, \quad \mathbf{H}(\mathbf{U}) = \begin{bmatrix} \rho w \\ \rho wu \\ \rho wv \\ \rho w^2 + p \\ (E + p)w \end{bmatrix}.$$

With these definitions, the conservative form of Euler system in $\mathbb{R}^3$ reads

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial z} = \frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = \mathbf{0}, \quad (6)$$

with state vector and flux tensor

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ E \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I} \\ (E + p)\mathbf{u} \end{bmatrix}, \quad (7)$$

where ρ is the density, $\mathbf{u} = (u, v, w)^\top$ the velocity, $E = \rho(e + \frac{1}{2}\|\mathbf{u}\|^2)$ the total-energy per unit volume and p the static pressure. This is the basis for the finite-volume discretisation implemented in OPENFOAM for the 3-D diamond-airfoil simulations. Note: $\|\mathbf{u}\| = \sqrt{u^2 + v^2 + w^2}$. In the quasi-2-D case ($w=0$) Eq. (6) reduces to the familiar 2-D expressions below:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0 \quad (8)$$

2.2 Equation of state and thermodynamic closure

The *Ideal Gas Law* is our equation of state that relates the variables, ρ which is our fluid density, p the local pressure, R the molecular gas constant, and T the local temperature.

$$p = \rho RT \quad (9)$$

Using the first law of Thermodynamics, p can be related to the specific internal energy e and the specific heat ratio γ by:

$$e = \frac{p}{\rho(\gamma - 1)} \quad (10)$$

These can be combined for a new expression that expresses p entirely in terms of the state vector $\mathbf{U}$, allowing the flux vectors to be evaluated directly from conserved quantities. This closes the system.

$$p = (\gamma - 1) \left(E - \frac{1}{2} \rho \|\mathbf{u}\|^2 \right) \quad (11)$$

2.3 Local Sound Speed and Mach Number

To analyze compressible flow behavior, we often need the local sound speed and Mach number. The local speed of sound and Mach number follow directly:

- Speed of sound, a :

$$a = \sqrt{\gamma p / \rho}, \quad (12)$$

- Mach number, M

$$M = \frac{\|\mathbf{u}\|}{a}. \quad (13)$$

2.4 Oblique-shock theory (θ - β - M relation)

For a sharp diamond airfoil the leading edges behave locally as two identical wedges that deflect the flow by a half-angle θ . The shock-wave angle β is obtained from the classical θ - β - M equation

$$\tan \theta = \frac{2 \cot \beta (M_1^2 \sin^2 \beta - 1)}{M_1^2 (\gamma + \cos 2\beta) + 2}, \quad (14)$$

where M_1 is the upstream Mach number. We always select the weak- shock root ($\beta < 90^\circ$). Once β is known the component of the Mach number normal to the shock can be calculated

$$\boxed{M_{n1} = M_1 \sin \beta}. \quad (15)$$

Equation (15) acts as a bridge: it converts the two-dimensional oblique-shock problem into an equivalent one-dimensional *normal-shock* problem, enabling the use of standard jump and isentropic relations.

2.5 Normal-Shock and Isentropic Relations (using M_{n1})

Normal-shock jumps

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n1}^2 - 1), \quad (16)$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_{n1}^2}{(\gamma - 1)M_{n1}^2 + 2}, \quad (17)$$

$$M_{n2}^2 = \frac{1 + \frac{1}{2}(\gamma - 1)M_{n1}^2}{\gamma M_{n1}^2 - \frac{1}{2}(\gamma - 1)}. \quad (18)$$

Total-pressure ratio across the shock

$$\frac{p_{02}}{p_{01}} = \left[\frac{(\gamma + 1)M_{n1}^2}{(\gamma - 1)M_{n1}^2 + 2} \right]^{\gamma/(\gamma-1)} \left[\frac{\gamma + 1}{2\gamma M_{n1}^2 - (\gamma - 1)} \right]^{1/(\gamma-1)}. \quad (19)$$

Isentropic relations (regions 1 and 2)

$$\frac{p_{01}}{p_1} = \left[1 + \frac{\gamma - 1}{2} M_1^2 \right]^{\gamma/(\gamma-1)}, \quad \frac{p_{02}}{p_2} = \left[1 + \frac{\gamma - 1}{2} M_2^2 \right]^{\gamma/(\gamma-1)}, \quad (20)$$

with

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)}. \quad (21)$$

Stagnation pressure at the apex

$$p_{\text{stag}} = p_{02} = \frac{p_{02}}{p_{01}} \frac{p_{01}}{p_1} p_1, \quad (22)$$

where the two ratios on the right-hand side are given by Eqs. (19) and (20), respectively.

2.6 Reference stagnation pressure for the baseline case

For the baseline configuration ($M_\infty = 3$, $\theta = 10^\circ$), Eqs. (14)–(19) give

$$\beta = 27.38^\circ, \quad M_{n1} = 1.38, \quad \frac{p_{02}}{p_{01}} = 0.96.$$

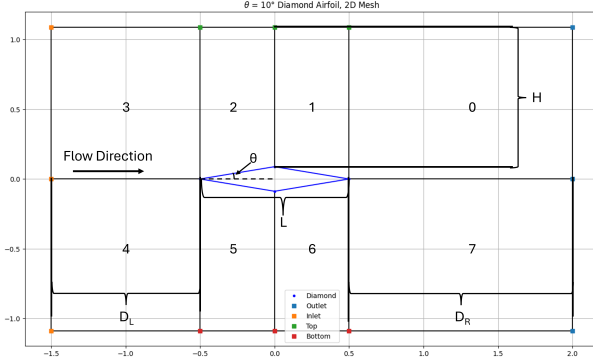
This value serves as the *theoretical target* against which the grid-refined 2-D CFD solutions will be validated before the three-dimensional extension.

3 Mesh Setup

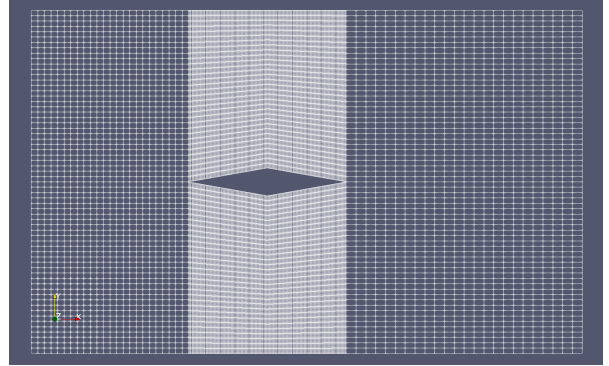
The mesh of the diamond-shaped wing is generated from *OpenFOAM*'s `blockMeshDict` which is set up as follows:

3.1 2D

In $\mathbb{R}^2$:



(a) 2D Mesh Schematic



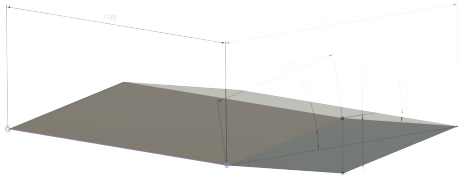
(b) 2D Mesh in *Paraview*

Where for our case:

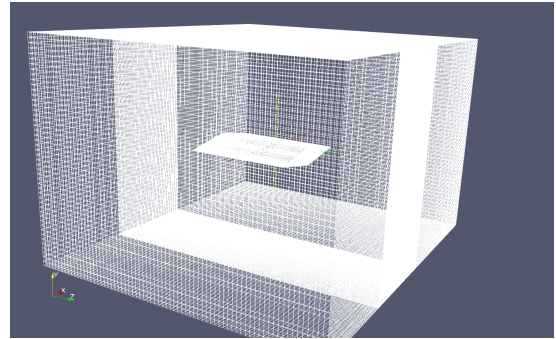
- $L = 1.0$ (Chord)
- $\theta = 10^\circ$ (Deflection Angle)
- $D_L = 1.0$ (Distance Left of the Diamond - Fore)
- $D_R = 1.5$ (Distance Right of the Diamond - Aft)
- $H = 1.0$ (Distance Above/Below the top/bottom-most point of the Diamond)

The size of the computational domain is then $(D_L + L + D_R) \times (2H + L \tan(\theta)) \times 0.1$. The `blockMeshDict` for the $\mathbb{R}^2$ case can be found in the **Appendix**.

3.2 3D



(a) CAD of Diamond Wing



(b) 3D Mesh in *Paraview*

We introduce a new term $D_Z = 1.0$ which is the extension of the computational domain from diamond wing in the z-direction. We also add two new boundaries for the front and back walls as `symmetryPlane`'s to close the 3D domain. The span of our diamond is $b = 1$. The `blockMeshDict` for the $\mathbb{R}^3$ case can be found in the **Appendix**. This `blockMeshDict` creates the bounding domain while Diamond-Shaped wing is made using CAD and the STL is meshed within the bounds using `snappyHexMesh`.

4 Mesh Refinement and Comparison to OST

In this section, we analyze three mesh resolutions: coarse, intermediate, and refined.

MeshID	Cells	Δt [s]
Coarse	16,896	0.004
Intermediate	67,584	0.002
Refined	1,081,344	0.0005

Table 1: Mesh and Solver Parameters

4.1 First Coarse Solution

For our coarse solution, we will visualize the properties pressure p , density ρ , velocity $\mathbf{U}$, and temperature T after simulating Mach 3 flow over the step for $T = 4$ s.

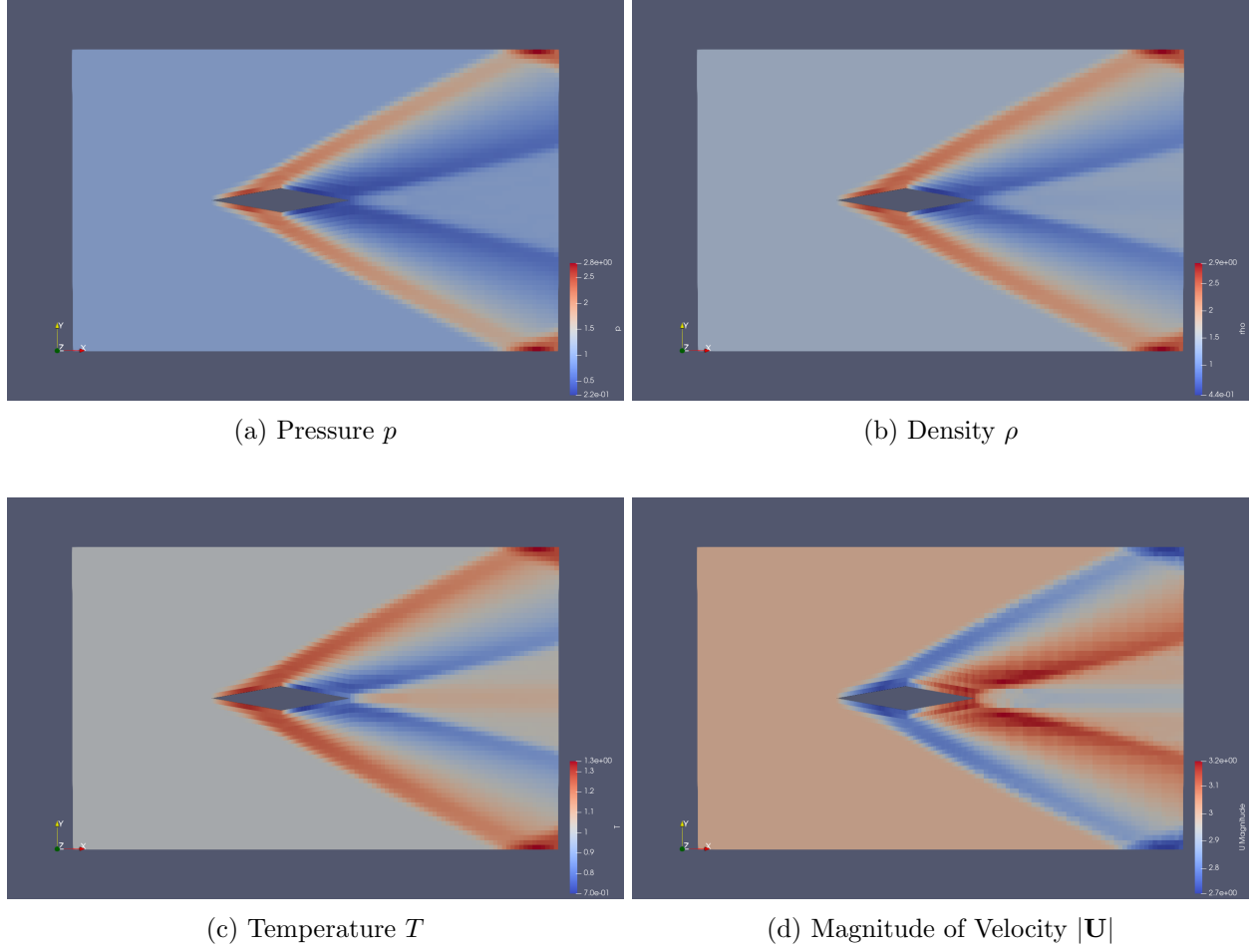


Figure 3: Properties of Coarse Mesh Solution

Using the built-in calculator in *Paraview*, we are able to employ Equations 12 and 13 to visualize the Mach and Speed of Sound in our supersonic simulation.

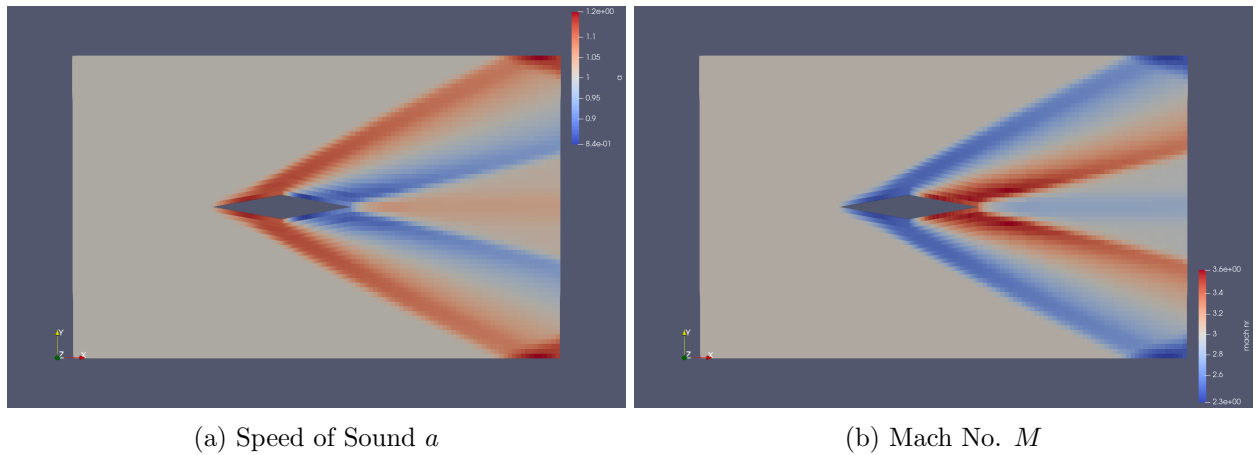
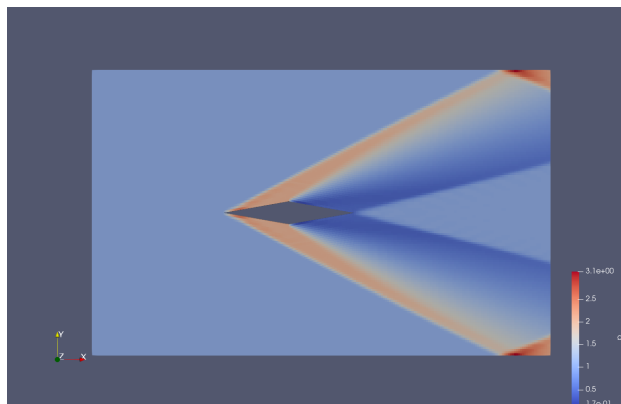
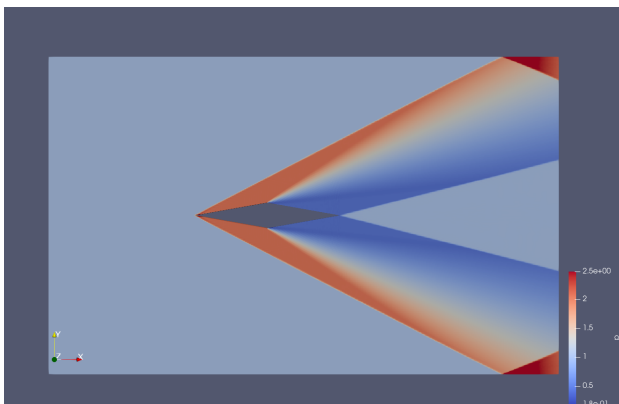


Figure 4: Mach and Speed of Sound of Coarse Mesh Solution

4.2 Intermediate and Refined Solution



(a) Intermediate Mesh Solution, Pressure p



(b) Refined Mesh Solution, Pressure p

4.3 Wave Angle and Pressure

Pressure along the forward facing top surface for all resolutions is shown in the Figures below.

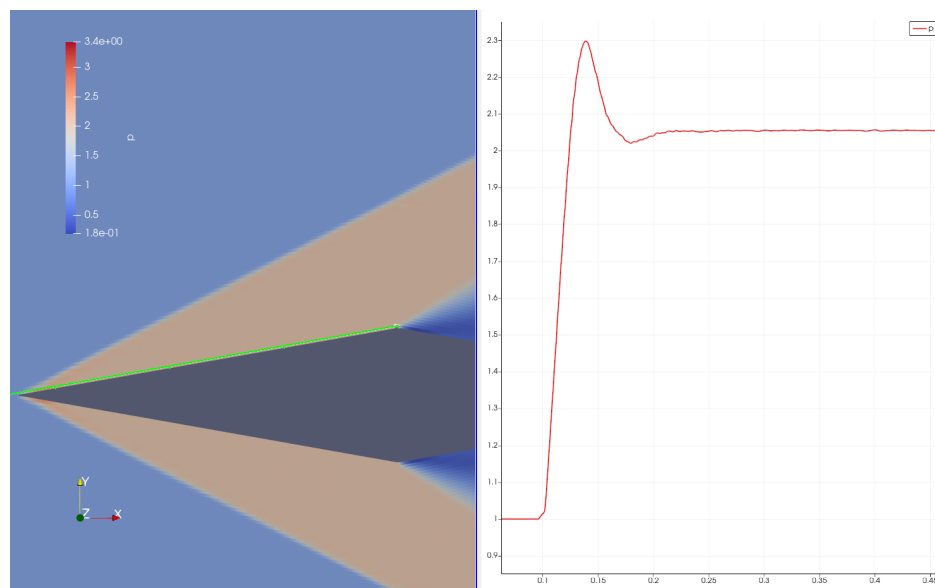


Figure 6: Pressure captured along the top front surface of the diamond wing section

To calculate wave angle, we place a vertical line at a distance $x = L/2$ downstream of the wedge apex (nose) and measure the shock height above the surface as y . Denoting by β the angle between the shock and the wedge surface, simple trigonometry gives

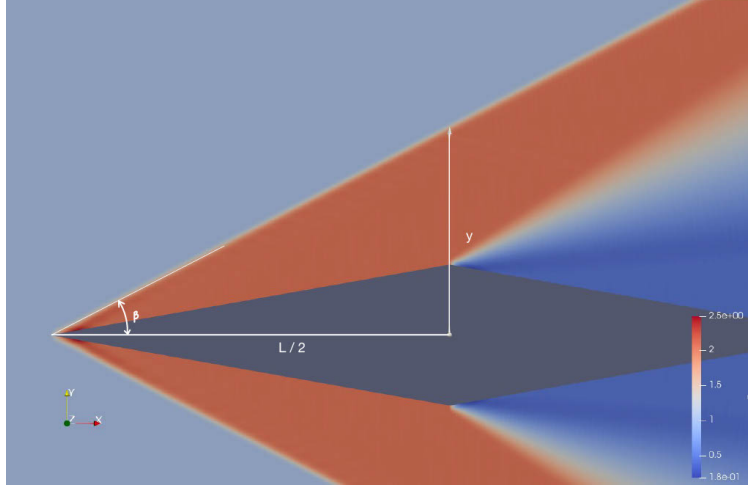


Figure 7: Calculation of Shock Angle Visual

$$\beta = \arctan\left(\frac{y}{L/2}\right) = \arctan\left(\frac{2y}{L}\right).$$

The oblique-shock angle β is measured with respect to the incoming flow direction, and the wedge deflection angle is $\theta = 10^\circ$. From the geometry (see Fig. 7),

$$\beta = \frac{y}{L/2} = \frac{2y}{L}$$

Table 2: Computed shock-wave angle β for different solution levels

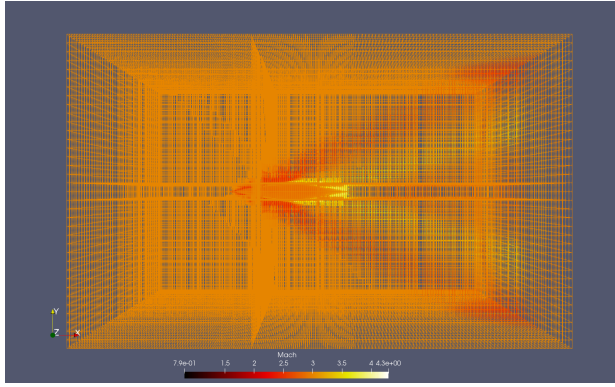
MeshID	y (m)	L (m)	β ($^\circ$)
Coarse	0.256	0.10	27.11
Intermediate	0.258	0.10	27.29
Refined	0.260	0.10	27.47
Theory			27.38

Table 2 compares the computed shock angles for three mesh resolutions against the theoretical value $\beta_{\text{theory}} = 27.38^\circ$. We define

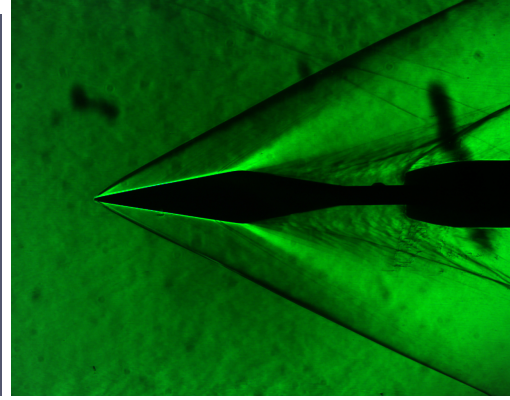
$$\Delta\beta = \beta_{\text{comp}} - \beta_{\text{theory}}, \quad \varepsilon = \frac{|\Delta\beta|}{\beta_{\text{theory}}} \times 100\%.$$

The coarse mesh underestimates the angle by $\Delta\beta = -0.27^\circ$ ($\varepsilon = 1.0\%$), while the intermediate mesh gives $\Delta\beta = -0.09^\circ$ ($\varepsilon = 0.33\%$) and the refined mesh $\Delta\beta = +0.09^\circ$ ($\varepsilon = 0.33\%$). Errors fall below 0.1° on the finer grids, indicating convergence toward the theoretical solution.

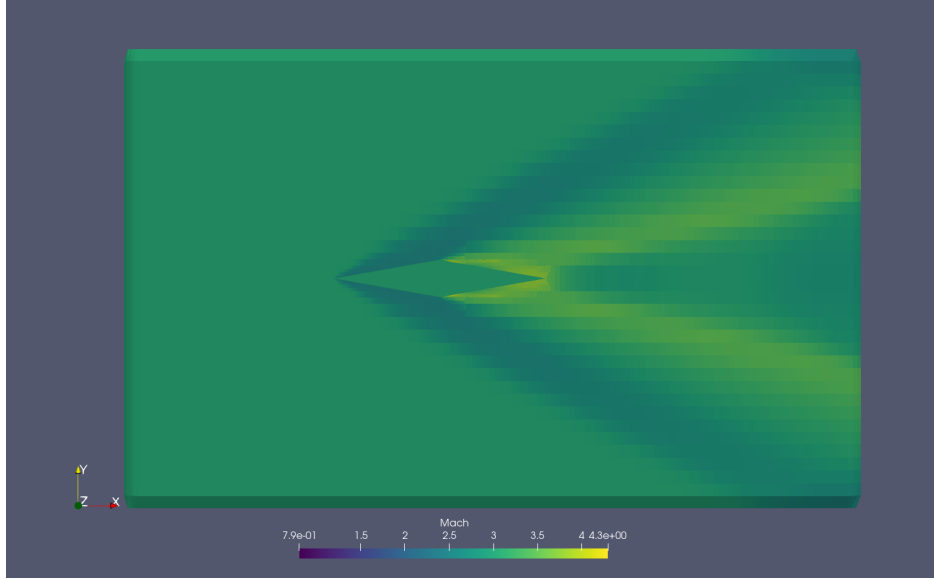
5 In Three Dimensions



(a) 3D Mesh in Wireframe view



(b) ASE 162M High-Speed Aerodynamics



(c) Slice of the 3D Result showing Mach No.

Figure 8: Comparison between the simulation and a real wind tunnel at Mach 3

A previous experiment involving a schlieren setup for visualization of supersonic flow in *High Speed Aerodynamics*, ASE 162M shows flow with a freestream Mach number $M_\infty = 3$ over the diamond airfoil at $\alpha = 0^\circ$, the experimentally measured shock-wave angle was

$$\beta_{\text{exp}} = 28.73^\circ,$$

while our 3D CFD simulation yielded

$$\beta_{\text{sim}} = 30.54^\circ,$$

an overprediction of $\Delta\beta = +1.81^\circ$ ($\approx 6.3\%$).

The discrepancy likely arises from three-dimensional flow expansion and residual numerical diffusion in the shock region. Despite the overestimate, the mesh resolution captures the overall 3D shock

structure well, and further local refinement or higher-order schemes could reduce the remaining error.

However, $\beta_{3D} \approx 30.54^\circ$, which is greater than the theoretical result from 2D Euler Equations, $\beta_{\text{theory}} = 27.38^\circ$. Due to 3D effects, the wave angle is expected to be greater given the flow relief in the spanwise direction. This turns out to be the case for this mesh.

The shocks and expansion fans are not as clear as they were in the 2D simulations. It appears transparent like vapour, which is a key visual aspect of a real supersonic shock!

Stats on this "coarse" mesh : $N_{\text{cells}} = 947084$, $N_{\text{faces}} = 3072728$, and $N_{\text{points}} = 1178825$.

6 Conclusion

A systematic mesh-refinement study in two dimensions demonstrated that the computed oblique-shock angle converges to within 0.1° of the theoretical value $\beta_{\text{theory}} = 27.38^\circ$, and the apex stagnation pressure agrees within 1 percent of theory—validating the mesh and solver fidelity for 2D inviscid flow.

When extended to a structured 3D mesh ($\approx \times 10^6$ cells), the predicted wave angle increases to

$$\beta_{\text{sim},3D} = 30.54^\circ,$$

moving the solution toward the experimental measurement $\beta_{\text{exp}} = 28.73^\circ$ and correctly capturing the characteristic steepening due to spanwise flow relief and boundary-layer effects. Although the absolute error relative to experiment is $+1.81^\circ$ (6.3%), the 3D simulation reproduces the observed trend that 2D Euler theory cannot.

In summary, while 2D inviscid theory and simulation agree with high precision, the 3D CFD model more faithfully reproduces experimental shock-wave behavior in finite-span diamond airfoils, underscoring the importance of three-dimensional effects in supersonic aerodynamics.

Appendix

A Code

Additionally, team repository: <https://github.com/andressuniaga/CFD-SP25-11>